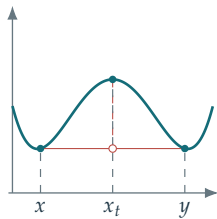


Panel 1

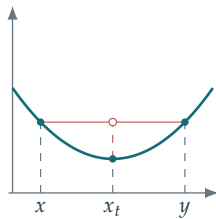
Nonconvex



$$f(x_t) > \ell_t$$

Panel 2

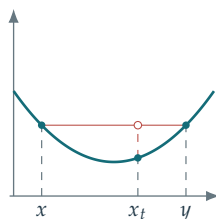
Convex



$$f(x_t) \leq \ell_t$$

Panel 3

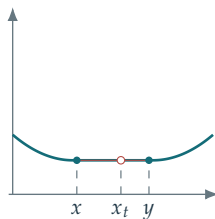
Strictly convex



$$f(x_t) < \ell_t$$

Panel 4

Convex, not strict



$$f(x_t) = \ell_t$$